Income and Adult Characteristics
Micro-Project 2

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Problem Statement

- Income differs across adults with different backgrounds and jobs.
- Question: Which demographic, educational, and employment characteristics occur together with earning more than $50,000 per year?
- The analysis looks at age, sex, education, occupation, and weekly work hours. Workclass and marital status add context in the notebook.
- The dataset gives two income categories, so the analysis compares the percentage above $50,000 within each group.

Hypothesis Formulation

- Higher income will be more common among adults with more education, middle-career ages, professional or managerial jobs, and longer workweeks.
- The differences will also appear when age, education, and work hours are considered together.
- Possible explanation: Education and experience may help people qualify for higher-paying jobs. Hours worked may also contribute to annual income.
- Grouped percentages and a stratified heatmap will explore these patterns. A classification model and formal hypothesis test come later.

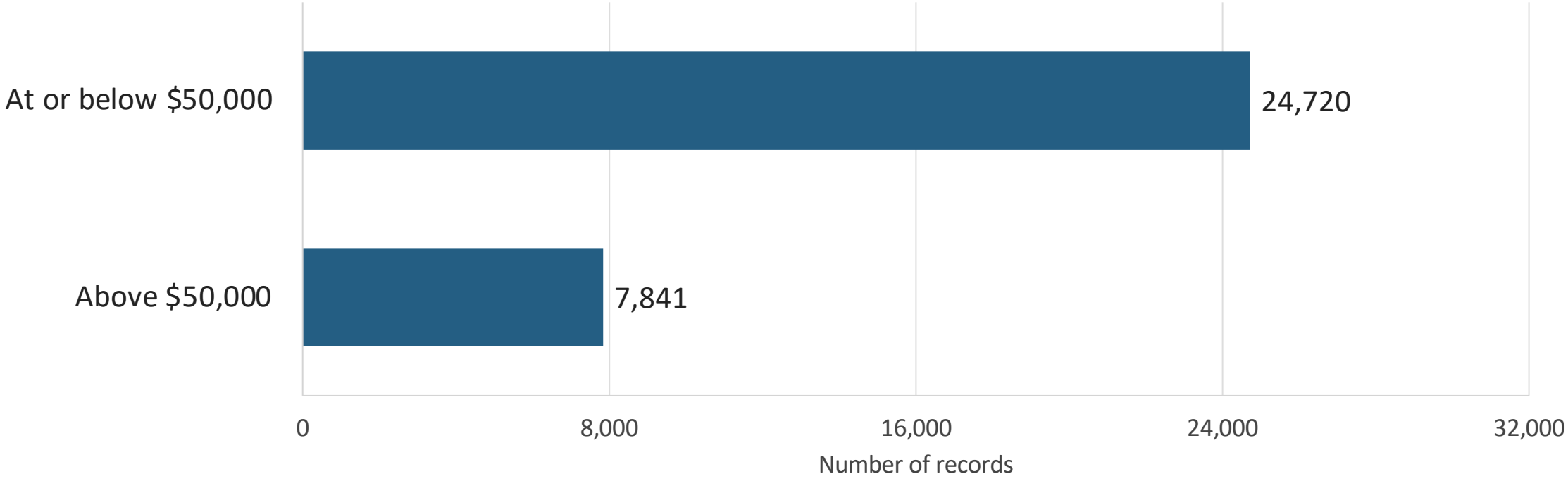
Acquire

- Source: UCI Adult dataset, based on the 1994 Census database.
- The supplied adult income.csv contains 32,561 records and 15 fields. It matches the training split. The separate test split is not used.
- Tools used: Jupyter Notebook, Python, Pandas, NumPy, Matplotlib, and Seaborn.
- Each row counts equally in these summaries. fnlwgt is the final survey weight, retained for later work.
- Reference: Becker & Kohavi (1996), Adult. https://doi.org/10.24432/C5XW20

Prepare

- Trim extra spaces, mark ? and blank values as missing, and convert the six numeric fields to integers.
- Check the field names, income labels, numeric ranges, and matching education codes. Keep all 32,561 records.
- Missing values: workclass 1,836, occupation 1,843, and native-country 583. Unknown occupation stays visible as a separate group.
- Keep 24 repeated profiles because the file has no person ID. Removing them changes the overall rate by only +0.012 percentage points.
- Create fixed age, education, and work-hour groups. Check each percentage against its numerator and group size.

Analyze: Income Categories

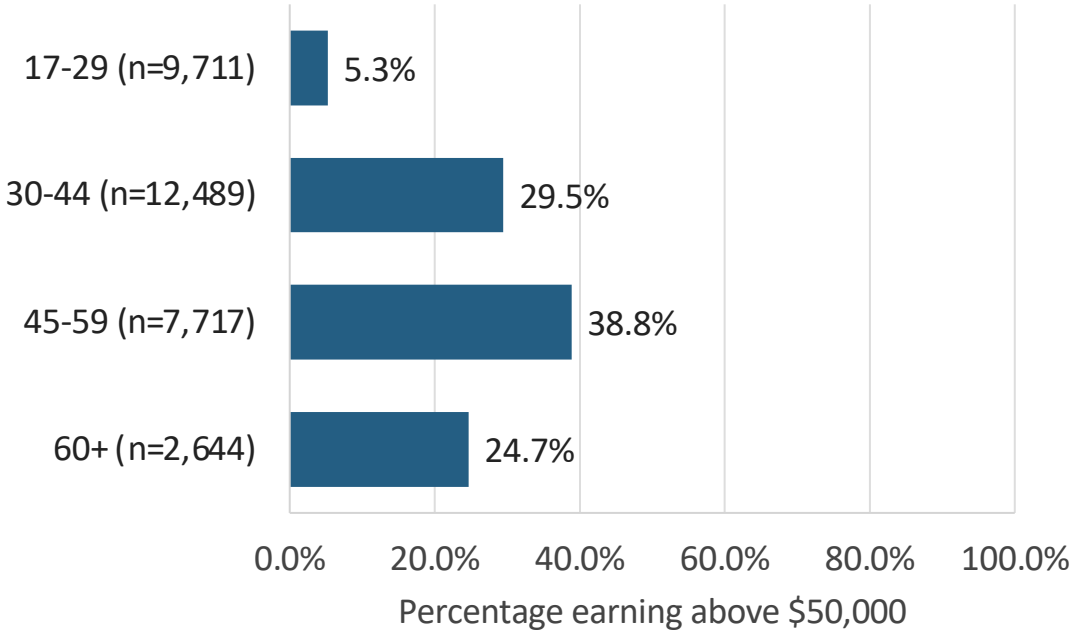


7,841 of 32,561 records earn above $50,000, or 24.1% of this sample.

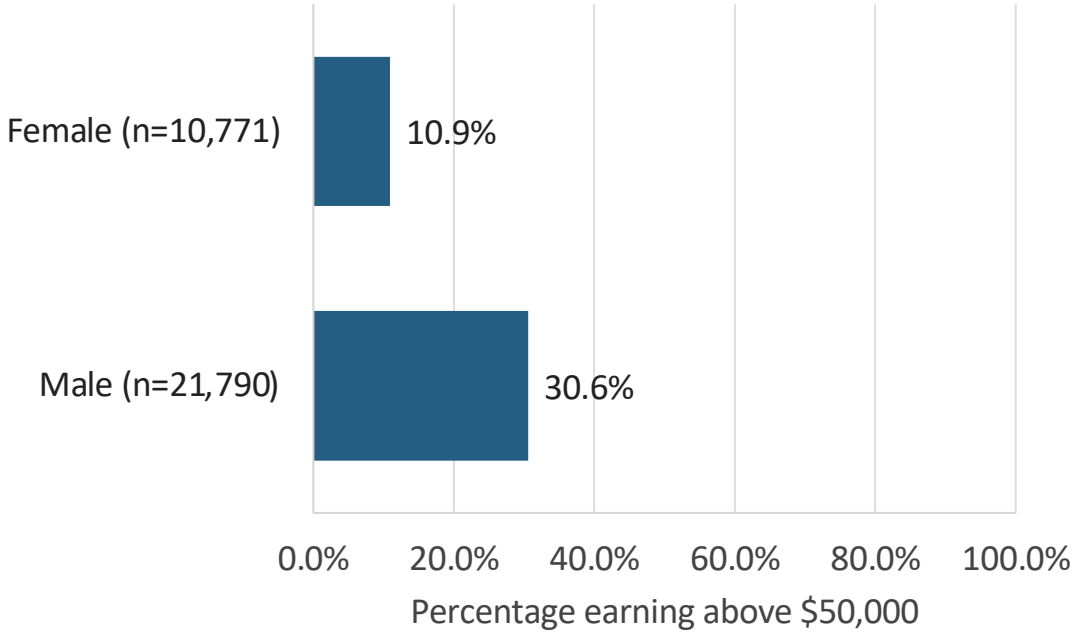
Source: UCI Adult, 1994 sample. Unweighted percentages. n = group size.

Analyze: Demographics

Age group



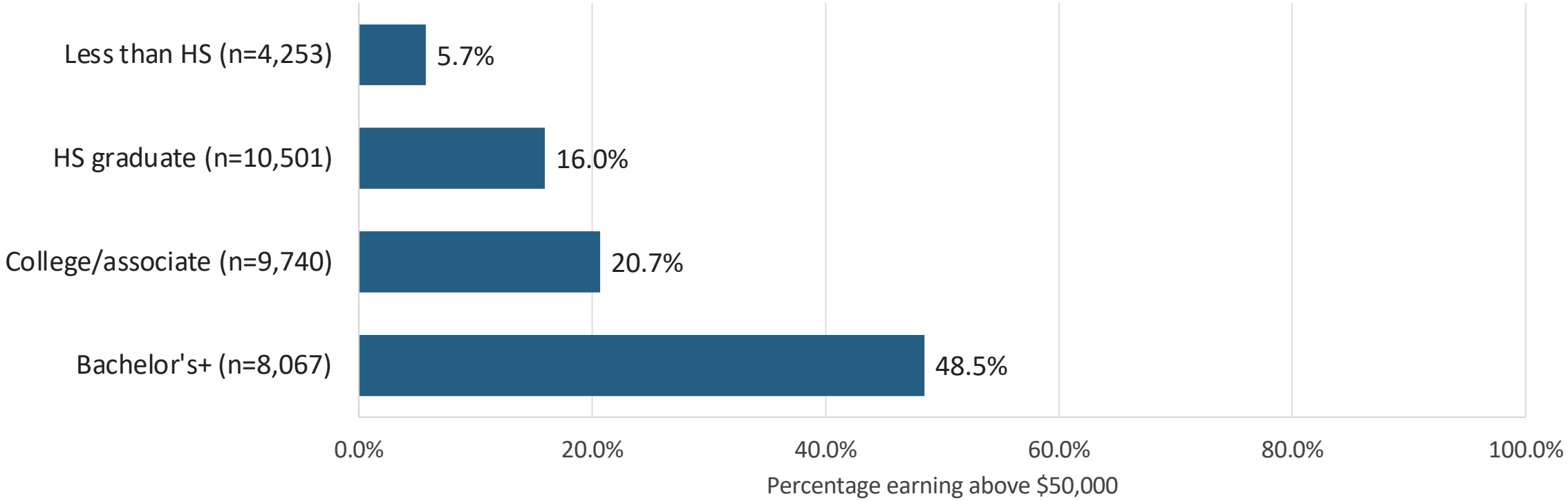
Sex recorded in the dataset



Ages 45-59 have the highest age-group rate (38.8%). Sex differences are unadjusted and may reflect other factors.

Source: UCI Adult, 1994 sample. Unweighted percentages. n = group size.

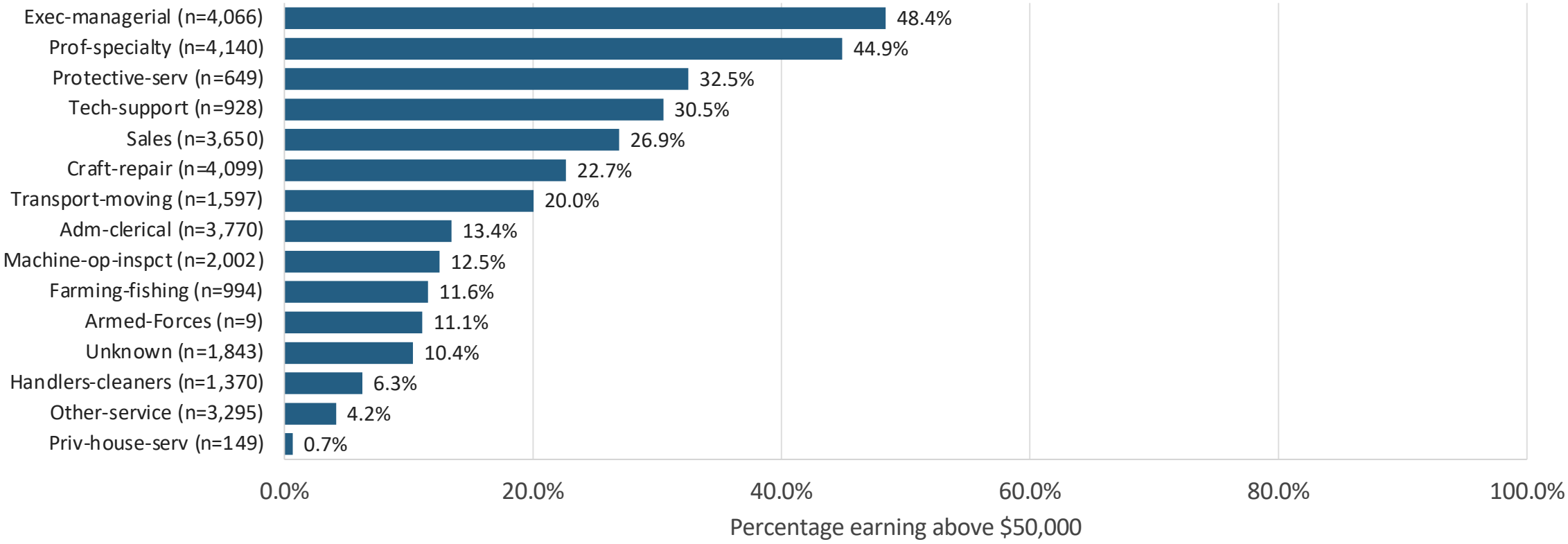
Analyze: Education



The rate is 48.5% for a bachelor's degree or higher, compared with 16.0% for high school graduates.

Source: UCI Adult, 1994 sample. Unweighted percentages. n = group size.

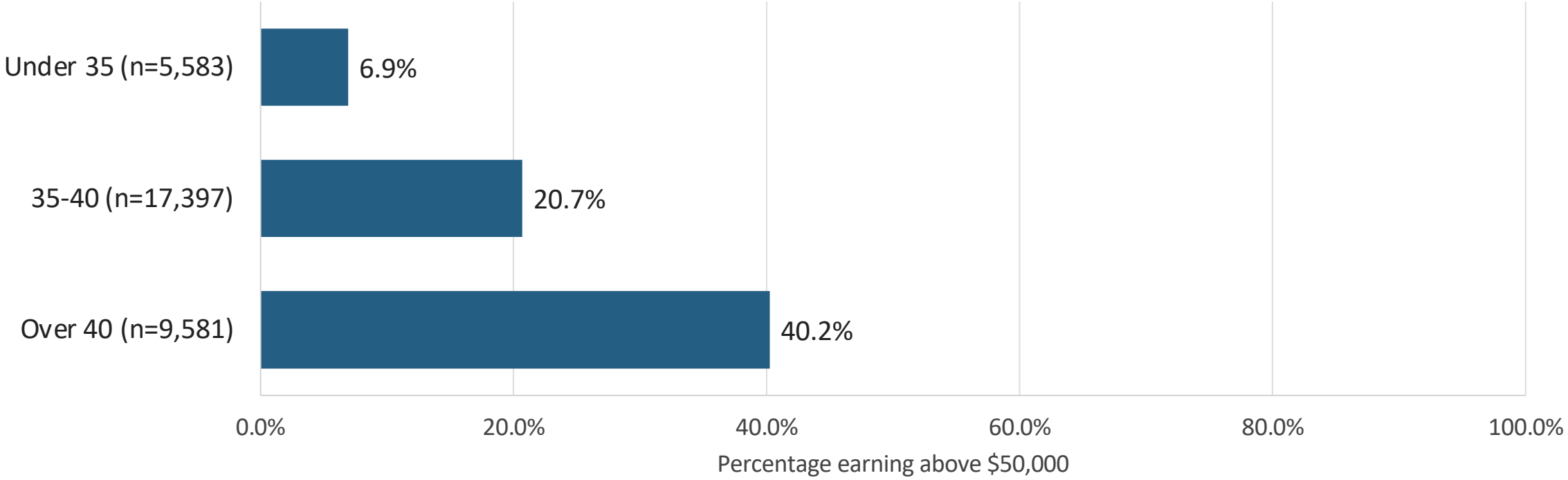
Analyze: Occupation



Executive/managerial and professional-specialty groups have the highest rates. Armed-Forces has only 9 records.

Source: UCI Adult, 1994 sample. Unweighted percentages. n = group size.

Analyze: Weekly Work Hours



The rate is 40.2% for adults working over 40 hours a week and 20.7% for those working 35-40 hours.

Source: UCI Adult, 1994 sample. Unweighted percentages. n = group size.

Analyze: Combined Characteristics

40 hours or less

Age	Less than HS	HS graduate	College/ associate	Bachelor's+
17-29	0.8% n=1,415	2.3% n=2,390	3.2% n=2,941	9.0% n=1,118
30-44	5.1% n=850	13.8% n=2,829	20.5% n=2,406	42.7% n=1,865
45-59	8.5% n=739	23.6% n=1,823	33.1% n=1,298	56.6% n=1,155
60+	6.0% n=499	13.2% n=760	22.2% n=459	44.1% n=433

Over 40 hours

Age	Less than HS	HS graduate	College/ associate	Bachelor's+
17-29	4.9% n=184	8.1% n=593	10.9% n=571	26.3% n=499
30-44	11.7% n=247	25.5% n=1,262	37.3% n=1,307	64.8% n=1,723
45-59	18.2% n=236	39.5% n=714	45.9% n=663	72.7% n=1,089
60+	n=83 small group	36.9% n=130	n=95 small group	68.1% n=185

Blue scale: 0-100%. Darker = higher percentage. Gray cells have fewer than 100 records.

Age 45-59, bachelor's or higher, and over 40 hours has a 72.7% rate (n=1,089), the highest displayed combination.

Source: UCI Adult, 1994 sample. Unweighted percentages. n = group size.

Report: Main Findings

- Education alone gives an incomplete picture. Age, occupation, and work hours also show substantial differences in the share above $50,000.
- Ages 45-59: 38.8%. Executive/managerial jobs: 48.4%. Professional-specialty jobs: 44.9%.
- The combined view adds detail. Among ages 45-59 with a bachelor's degree or higher, the rate is 56.6% at 40 hours or less and 72.7% above 40 hours.
- These descriptive patterns are consistent with the hypothesis. They do not establish each variable's independent effect or prove the explanation.

Report: Limits and Next Analysis

- The data describe a historical 1994 sample and broad income categories. They do not measure current salaries.
- The percentages are unweighted. Missing occupations and very small groups limit some comparisons.
- Group differences may reflect several related characteristics. This analysis does not establish causation or predictive accuracy.
- Later classification work can examine variables together, with a separate evaluation set and clear handling of missing values.
- This micro-project ends at Step 4, Report.